RelCompat3D: Re-Ranking 3D Scene Graph Relations with Geometric Evidence Supplementary Material

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Supplementary Method and Reproducibility Details

Overview. This supplement is organized around the evidence needed to assess and reproduce the main claims. We first give notation, target-construction rules, formal properties, estimator details, and the row-level regeneration check. We then report additional ablations, alternative-geometry audits, score and routing sensitivities, statistical intervals, transfer results, family-level analyses, and qualitative examples. Compact tables retain the comparisons that change the interpretation of the method. Full per-condition rows used to generate these summaries are included in the accompanying code and data package.

Notation

Table 1 summarizes the notation used in the main paper and the supplement.

Symbol	Meaning
r_i	Relation candidate i
(s_i, o_i)	Ordered subject and object instance identifiers
$p_i,\ a_i$	Predicate label and relation-family label
$k_i^{\mathrm{pair}},\ k_i^{\mathrm{rel}}$	Ordered-pair identity and exact relation-candidate identity
$T_i,\ G_i,\ Z_i$	Predicate semantics, ordered-pair measurements, and source relation score
$q,\ f_q,\ C_i^q$	Compatibility estimator, compatibility logit, and bounded compatibility score
$H_{a_i},\ \mathcal{O}_i$	Applicable transformation group and transformation orbit
$C_i^{\mathrm{tr,q}},\ u_i^q$	Transformation-consistent compatibility and within-family ranking score
$\mathcal{A}_{\mathrm{eval}},\ \mathcal{A}_{\mathrm{rank}}$	Evaluated and re-ranked relation-family sets
$\pi^Z,\ \pi_a^q$	Source ranking and ordered list for family a
$L_K(c),\ Y_c$	Selected top-K candidates and exact ground-truth relations in context c
$D_K,\ N_s, N_u, N_v$	Status-assigned candidates and satisfied, uncertain, and violated counts
$\mathcal{P},\ \ell_i^q$	Linked training pairs and compatibility logit used by the pairwise loss

Table 1: Notation used in the RelCompat3D method and evaluation.

Predictor	Candidates	Minimum	Maximum
VL-SAT	110,424	5.30×10^{-22}	0.9954
Open3DSG	79,722	0.6394	0.9281
SGFN	110,424	4.61×10^{-20}	0.5846

Table 2: Observed ranges of source relation scores for candidates in the re-ranked proximity and vertical-order families. No score is negative or exactly zero.

Model and Preprocessing Details

Open3DSG. Open3DSG uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`, selected using training and development losses. Public preprocessing produces candidates for 533 of the 548 contexts. The main evaluation retains all 548 contexts and treats the 15 contexts without candidates as empty sets. An eligible-context sensitivity analysis uses 3,899 relations. A separate recovery covers all 548 contexts by lowering the minimum number of visible objects to two and regenerating views for two scans.

SGFN. We use the released `SGFN_full_l160` model from the SceneGraphFusion 3DSSG implementation, with its 160-object, 26-relation checkpoint on the same evaluation scans and denominator.

Observed ranges of source relation scores. Table 2 reports the source relation scores observed among the proximity and vertical-order candidates evaluated by the primary re-ranking procedure. All scores are nonnegative, so multiplication by compatibility does not reverse the source-score ordering through a sign change. These are observed evaluation ranges rather than calibrated probability intervals. Scores are also nonnegative before family-aware re-ranking for Open3DSG. Its range over all candidates is $[0.5772, 0.9707]$.

Compatibility Estimation and Family-Aware Re-Ranking

Guarantees and Target Construction.

Proposition 1 (Exact transformation consistency) *Let a finite transformation group H_a act jointly on the predicate and geometry of an ordered pair, and define $C_a^{\mathrm{tr}}(x) = |H_a|^{-1} \sum_{g \in H_a} C_a(gx)$. Then $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$ for every $h \in H_a$.*

Proof. For any $h \in H_a$, $C_a^{\mathrm{tr}}(hx) = |H_a|^{-1} \sum_{g \in H_a} C_a(ghx)$. Right multiplication by h is a bijection on the finite group, so re-indexing $g' = gh$ gives $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$. The two-element proximity and vertical actions yield the identities in the main paper, and the singleton support/contact action is the identity.

Family-sequence preservation. For each family a, the algorithm constructs the ordered list π_a defined in the main paper. At position j, it selects the next unselected candidate from π_{a_j}. The output family sequence is therefore exactly $(a_1, \ldots, a_n)$. The support/contact list follows the source ranking, so the first K positions contain exactly the same ordered support/contact subsequence as the source prefix. Therefore the support/contact selections and their order are identical at every reported K.

Prefix utility optimality. Fix a prefix length K, and let $n_a(K)$ be the number of positions assigned to family a in the source prefix. Every feasible ranking that preserves these family counts must select $n_a(K)$ candidates from each family. Because each re-ranked π_a is sorted by u_i, each estimator selects the $n_a(K)$ highest-scoring candidates in that family. Replacing one of these candidates with a lower-scoring candidate from the same family would reduce the summed score of the prefix. The support/contact subsequence is fixed, so it does not affect this comparison.

Counterfactual target construction. Ground-truth relations in the evaluated families provide positives. Negatives are generated within the same scene context and linked to their source positive. The construction does not use source relation scores, predictor identities, or evaluation candidates. Table 3 gives the complete construction policy.

Information-use boundary. Table 4 makes the information boundary explicit. Training examples come only from the training split, so evaluation candidate rows, source relation scores, and primary verifier-status labels do not enter target construction. The counterfactual rules and primary verifier nevertheless share some OBB-derived measurements and family thresholds. The point- and mesh-based audit removes that OBB overlap but retains the reconstructed evaluation scenes and relation ontology.

We cap negatives at two per positive, 200 per context–family, and three times the number of positives per family. These caps limit repeated easy negatives but do not equalize family counts. The constructed training and internal-development sets together contain 28,977 positives and 37,477 negatives. Their combined family totals are 23,008 proximity, 5,488 vertical, and 37,958 support/contact rows. The explicit overlap above is why Violation computed by the rule-based geometry verifier is reported as a verifier-derived measure rather than independent physical-validity ground truth. The feature-removal analysis tests this dependence but does not eliminate it.

Family	Intervention and acceptance rule	Relation to verifier
Proximity	Choose directed same-context pairs in descending normalized XY distance. Require distance $\geq$ 2.5 and zero projected overlap.	Shares distance/overlap primitives and the 2.5 satisfied boundary. The verifier declares violation only at $\geq$ 3.5.
Vertical	On the same directed pair, replace `higher than` with `lower than` or conversely. Require the positive direction to have $\lvert\Delta z\rvert \geq 0.25$ m and normalized $\lvert\Delta z\rvert \geq 0.15$.	Shares both signed primitives and thresholds.
Support/ contact	Replace the subject or object within the context. Reject known support/contact pairs in either direction and floor subjects. Require any two of normalized XY distance $\geq$ 2.0, zero overlap, and vertical gap ≥ 0.30 m.	Shares these OBB primitives and thresholds. Point-subtype evidence used by the reported verifier is not used to generate negatives.

Table 3: Counterfactual-negative construction. Normalized XY distance uses the mean ordered-pair XY diagonal. Zero overlap requires both directed projected-overlap ratios to be at most 10^{-9}.

Estimator and Optimization Details. RelCompat3D-Linear fits one logistic model per evaluated relation family. Numeric features use means and standard deviations estimated from the training split. Missing values use the corresponding training mean. Inputs are predicate indicators T, predicate-independent geometric measurements of the ordered pair G, and the two predicate-signed height interactions defined in the main paper. The family label selects the head and is not repeated as an input. The source relation score, source rank, predictor identity, and object class are excluded. Transformation-consistent augmentation preserves the proximity predicate under endpoint swap. Vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

RelCompat3D-MLP fits one shared multilayer perceptron (MLP) with a single two-unit ReLU hidden layer and a predicate-linear skip connection. Its 29 inputs comprise family and predicate indicators, the same 17 geometric measurements, the two signed-height interactions, and the sum of the directional overlap ratios. The family indicator is nonconstant because this estimator shares parameters across families. It has 69 parameters and excludes the source relation score and predictor identity.

The 60,208 training rows yield 33,961 linked pairs, and

Information	Training construction	Primary verifier	Point/mesh audit
Evaluation candidate rows	No	Yes	Yes
Source relation score	No	No	No
Primary verifier-status labels	No	Output	No
OBB-derived measurements	Some	Yes	No
Point/mesh measurements	No	No	Yes
Evaluation scene identities	No	Yes	Yes
Relation ontology	Yes	Yes	Yes

Table 4: Information used by target construction and the two evaluation methods. "Some" denotes the OBB primitives and family thresholds shared by counterfactual construction and the primary verifier. The point- and mesh-based audit uses alternative measurements, not independent physical-validity ground truth.

transformation-consistent augmentation produces 86,032 optimization rows for both estimators. Linear uses 800 deterministic full-batch gradient-descent steps at learning rate 0.2. MLP uses 120 full-batch Adam steps at learning rate 0.02. Both use the same pairwise weight, margin, ℓ_2 coefficient, and fixed seed policy. The three linear family models contain 21–23 parameters. Each reported condition uses one fit with a fixed seed. The paired intervals quantify scan resampling rather than variation across training runs.

Additional Experiments

Experimental Setup

Implementation and Development Diagnostics. All preprocessing, fitting, and evaluation jobs ran in the Docker environment described in the code package. Compatibility is computed once for each candidate and cached before reranking. Across five timed runs over all 548 contexts, end-to-end re-ranking took 1.81–2.45 seconds per predictor, or at most 4.48 ms per context.

Development-split checks were diagnostic and did not select predictor-specific models. The Linear and MLP estimators obtained overall AUROC/Brier values of 0.9973/0.0157 and 0.9904/0.0225, respectively. Both satisfied the proximity and vertical-order transformation identities to numerical precision. The MLP ordered 99.23% of linked positive–counterfactual development pairs correctly. These checks support optimization and implementation validity. The paper claims are based on the fixed validation protocol.

Row-Level Regeneration Check. We create a deterministic set of derived rows for the 548-context evaluation universe. It contains pseudonymized candidate and context identities, predictor, predicate and family labels, source relation score, Linear and MLP compatibility, paper ranking positions, exact-match membership, and primary and point- and mesh-based verifier statuses. Original scan and instance identifiers, object categories, and raw geometric measurements are excluded. From these rows, one Docker command regenerates Tables 1–3 and the data for Figure 3. It also creates a verification rendering and compares every cell with the reported paper values. The completed run checks 291 cells. The maximum absolute error is zero under the 10^{-12} tolerance. The anonymous code package includes the exporter, reproduction script, schema, compact outputs, and hash manifests. Distribution of the derived rows depends on the dataset terms.

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Linear full	.9277/.0197	.9658/.0295
	Linear no pairwise	.9277/.0198	.9658/.0297
	Linear no averaging	.9277/.0197	.9660/.0295
	MLP full	.9272/.0189	.9650/.0296
	MLP no pairwise	.9275/.0180	.9648/.0280
	MLP no averaging	.9275/.0189	.9645/.0296
Open3DSG	Linear full	.4418/.0342	.5685/.0324
	Linear no pairwise	.4418/.0342	.5692/.0324
	Linear no averaging	.4413/.0342	.5692/.0324
	MLP full	.4670/.0413	.5989/.0371
	MLP no pairwise	.4600/.0399	.5939/.0362
	MLP no averaging	.4507/.0398	.5735/.0368
SGFN	Linear full	.7450/.0263	.9303/.0350
	Linear no pairwise	.7450/.0264	.9303/.0353
	Linear no averaging	.7450/.0263	.9303/.0350
	MLP full	.7457/.0258	.9288/.0350
	MLP no pairwise	.7452/.0255	.9295/.0330
	MLP no averaging	.7457/.0258	.9300/.0352

Table 5: Matched component removals. Cells are exact-match Recall / verifier-derived Violation reported as proportions. Full, no-pairwise, and no-averaging conditions are matched separately within each estimator.

The same rows can instead be regenerated from licensed external inputs.

Results

Component and Feature-Removal Analyses.

Matched component diagnostics. Tables 5–6 isolate the linked pairwise loss and transformation averaging for both compatibility estimators. Within each estimator, all conditions retain the same rows, features, optimizer, source relation scores, and family-aware re-ranking procedure. The No pairwise condition removes only the linked pairwise term. It retains relation-preserving augmentation and transformation averaging. The No averaging condition uses the fitted full estimator without inference-time transformation averaging.

The pairwise term raises the held-out positive-win rate by 0.085 percentage points for Linear and 0.057 points for MLP. It also increases the mean Linear margin and reduces its softplus margin loss from .03607 to .03534. The MLP margin summaries are mixed: its mean margin and softplus loss do not improve. The term is therefore treated as a training regularizer with limited, estimator-dependent diagnostic effect, not as the main cause of the aggregate Recall–Violation changes.

Transformation averaging gives zero compatibility error and identical top-K membership under transformed representations for both full estimators and their no-pairwise refits. Without averaging, the minimum top-K Jaccard is .9635 for Linear and .6975 for MLP. Thus averaging supplies the claimed exact consistency guarantee even though its aggregate effect in Table 5 is small.

Training-seed robustness. Five model seeds were fixed before training while the constructed rows and linked-pair identities were held constant. The reported MLP fit was included but not selected from this analysis. Linear is deterministic and produces the same model hash in all five runs. MLP variation is small. All predictor–K cells retain Recall no lower and Violation no higher than Source except one VL-SAT run at $K = 50$. That run selects one fewer exact relation than Source (92.699% versus 92.724% Recall) while retaining lower Violation (1.821% versus 2.675%). Complete per-seed means and standard deviations are included in the machine-readable results.

The factor ablation uses the same split and estimator class in a separate pooled analysis. Because one head spans all families, its semantic input includes predicate and family indicators. On the development split, semantic-only, predicate-independent G-only, additive semantics+G, and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. Shuffled and wrong-pair geometry reduce the compatibility signal. These results confirm that the estimator uses both factors and satisfies the implemented transformation identities. They do not turn the constructed target into independent physical-validity ground truth.

Table 7 reports the feature-removal results under the same family-aware re-ranking procedure.

Every feature-removal model is refitted on the 1,061 training scans with the same linked positive–counterfactual objective, exclusion of the source relation score, and exact transformation averaging. The exact-scalar condition retains development AUROC of .99999 for proximity and .99794 for vertical order. Removing all related distance or vertical measurements lowers these values to .9512 and .7089. Alternative evidence yields .8763 and .5531. At $K = 50$, the strict conditions retain a clear Open3DSG improvement but not the full Violation reduction on VL-SAT or SGFN. At $K = 100$, all feature-removal conditions improve both reported metrics across all three predictors. These results show that performance does not depend on one verifier input alone. They do not eliminate the broader dependence on correlated geometry. The main paper reports matched controls under family-aware re-ranking. The pooled family-conditioning result is summarized below.

Point- and Mesh-Based Consistency Audit. The primary verifier and compatibility estimators share some OBB-derived measurements. We therefore construct an alternative audit from reconstructed instance points and meshes. Vertex-based distances and heights provide the point estimate. Area-weighted samples from reconstructed mesh triangles provide the mesh estimate. A candidate is labeled satisfied or violated only when these estimates agree. A disagreement is labeled uncertain. Thresholds are fixed from training-split positives. Neither estimate uses OBB measurements, source scores, learned compatibility, predictor identity, or primary verifier labels.

Table 8 reports the agreement-based Violation values for both estimators at all five K values.

For each estimator, agreement-based Violation is lower than or tied with Source in all 15 predictor–K settings. The only tie is SGFN at $K = 5$. The paired 95% interval is below zero in every non-tied setting. Point-only and mesh-only point estimates have the same non-increasing direction. At $K = 50$, measured coverage ranges from 95.5% to 98.3%, and decidable coverage ranges from 71.0% to 85.7% across predictors and estimators. Full point-only, mesh-only, coverage, and interval rows are included in the machine-readable results.

Synthetic point and mesh interventions further check the audit implementation. Translating one object laterally changes proximity labels as expected, and reversing the relative heights changes vertical-order labels. These checks are implementation diagnostics rather than additional evidence of physical validity.

Qualitative Pair Analysis. The Open3DSG context used for the main-paper teaser also contains a complementary promotion case. The exact-label relation "desk close by chair" is verifier-satisfied, with an XY center distance of $0.436\,\mathrm{m}$. RelCompat3D-Linear moves this candidate from source rank 81 to rank 30, bringing it into the top-50. This case records a promotion outcome without implying a direct exchange with the vertical-order candidate, because the two candidates belong to different relation families.

Figure 1 links each ordered-pair projection to the measured evidence and ranking outcome. The proximity and vertical-order examples are demoted by both compatibility estimators. The support/contact example is retained in source order because that family is outside the re-ranking scope.

Counterfactual-Policy Sensitivity. We refit the Linear estimator after varying one construction or objective parameter at a time. The tested values are 2.0 and 3.0 m for the proximity threshold (default 2.5 m), 0.20 and 0.30 for the vertical margin (default 0.25), 1 and 4 for the negative cap (default 2), and 0.125 and 0.5 for the pairwise-loss weight (default 0.25). Across these eight perturbations, linked positive–counterfactual ordering remains perfect. At $K = 50$, the maximum absolute change from the active Linear model is 0.20 percentage points in Recall and 0.11 points in Violation. At $K = 100$, the corresponding maxima are 0.35 and 0.20 points. Every condition retains Recall no lower and Violation no higher than Source. Full rows are provided with the experiment artifacts.

Estimator, Fusion, and Routing Sensitivities.

Source-score mapping sensitivity. The source ranking is invariant to a strictly monotonic score mapping, whereas

Estimator	Condition	Family	Mean	P95	Max	Min Jaccard	Min exact-context
Linear	Full	Both	.0000	.0000	.0000	1.0000	1.0000
Linear	No pairwise	Both	.0000	.0000	.0000	1.0000	1.0000
Linear	No averaging	Proximity	.0150	.0817	.4532	.9635	.8266
Linear	No averaging	Vertical	.0014	.0080	.0192		
MLP	Full	Both	.0000	.0000	.0000	1.0000	1.0000
MLP	No pairwise	Both	.0000	.0000	.0000	1.0000	1.0000
MLP	No averaging	Proximity	.1684	.5805	.9370	.6975	.4617
MLP	No averaging	Vertical	.0402	.2896	.9157		

Table 6: Transformation diagnostics. Mean, P95, and Max are the worst source-specific summaries of the absolute compatibility difference under the applicable endpoint/predicate transformation. The last columns give the minimum transformed-view top-K membership consistency over all predictors and reported K.

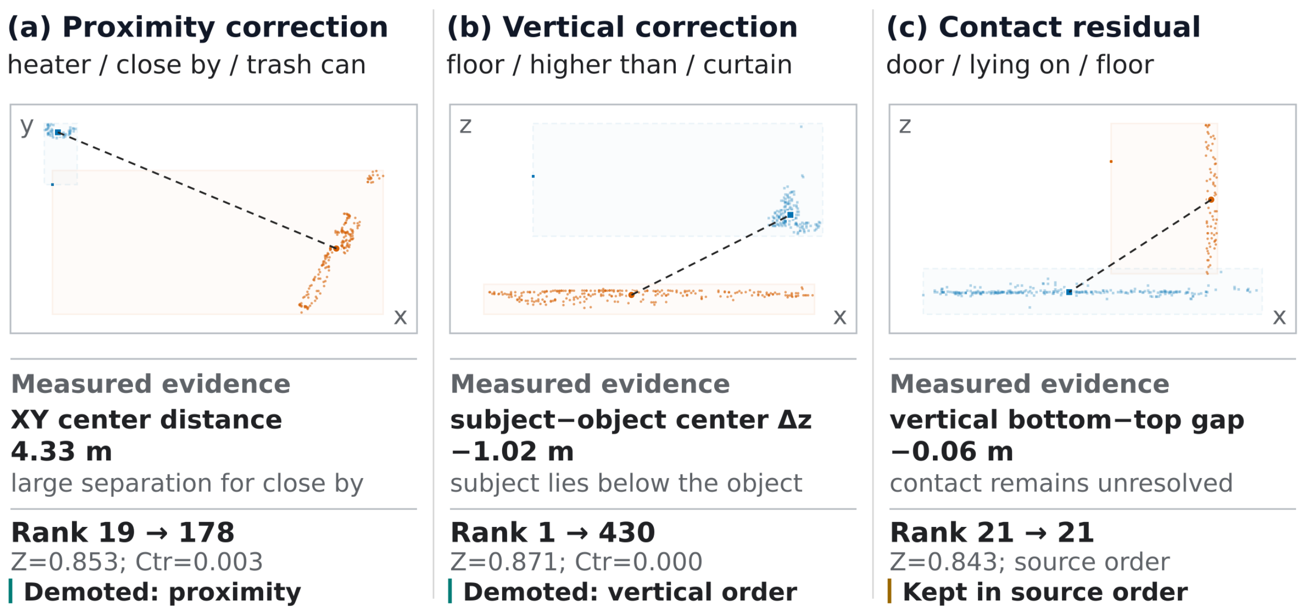

Figure 1: Pair–evidence–outcome analysis. Orange circles and solid boxes identify the subject. Blue squares and dashed boxes identify the object. The first two violations are demoted. The support/contact example requiring contact evidence is kept in source order.

the product $Z_i C_i^{\mathrm{tr,q}}$ need not be. We therefore apply a fixed grid of mappings before product scoring without selecting a replacement mapping from evaluation results. The smooth grid contains

$$g_\gamma(Z) = Z^\gamma, \quad \gamma \in \{0.5, 2, 4\},$$

and

$$g_\tau(Z) = \sigma\big(\mathrm{logit}(\bar{Z})/\tau\big)\,,$$
$$\bar{Z} = \mathrm{clip}(Z, \epsilon, 1-\epsilon),$$
$$\epsilon = 10^{-6}, \qquad \tau \in \{0.5, 2\}.$$

The temperature mappings are monotonic scale stresses and do not treat the Open3DSG cosine score as a calibrated probability. A separate context-and-family percentile mapping assigns descending scores from one to zero within each context and family.

Table 9 summarizes all five reported K values. A setting meets both criteria when Recall is no lower and verifier-derived Violation is no higher than Source. All 75 Linear settings and 74 of the 75 MLP settings meet both criteria under the smooth mappings. The only exception is VL-SAT MLP at $K = 50$ under Z^4, where Recall changes by -0.025 percentage points, its paired interval includes zero, and Violation decreases. The percentile stress produces two Linear and four MLP Recall losses, none larger than 0.227 percentage points, without increasing Violation. Across the complete grid, including percentile mappings, the minimum pair-weighted Kendall correlation with identity-product ranking is 0.810 and the minimum $K = 50$ micro-Jaccard overlap is 0.815. These results show limited sensitivity over the tested mappings, not score-scale invariance.

Closest simple baseline. The non-learned robust-density baseline is fitted only from training-split positive geometry. It uses predicate-specific medians and interquartile ranges, transformation averaging, the same product score, and the

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Source	.9272/.0268	.9635/.0476
	Full model	.9277/.0197	.9658/.0295
	Exact verifier scalar removed	.9280/.0199	.9660/.0300
	Related measurements removed	.9275/.0268	.9653/.0468
	Alternative evidence only	.9280/.0268	.9642/.0471
Open3DSG	Source	.4043/.1387	.5111/.1242
	Full model	.4418/.0342	.5685/.0324
	Exact verifier scalar removed	.4381/.0342	.5687/.0325
	Related measurements removed	.4119/.0954	.5264/.0955
	Alternative evidence only	.4104/.1194	.5327/.1188
SGFN	Source	.7402/.0385	.9235/.0630
	Full model	.7450/.0263	.9303/.0350
	Exact verifier scalar removed	.7465/.0268	.9300/.0358
	Related measurements removed	.7467/.0388	.9298/.0616
	Alternative evidence only	.7462/.0386	.9277/.0625

Table 7: Feature-removal analysis. Cells are exact-match Recall / verifier-derived Violation under the same family-aware re-ranking procedure. Exact verifier scalar removed omits the normalized scalar consumed by the verifier. Related measurements removed omits all directly related distance or vertical measurements. Alternative evidence only retains projected overlap for proximity and horizontal-distance/overlap context for vertical order.

same family-aware re-ranking rule. It does not use verifier labels from the evaluation or constructed counterfactual negatives. Table 10 reports all K values. At $K = 50$, both learned estimators have higher Recall and lower Violation than this baseline for all three predictors. Over all 15 predictor–K settings, Linear improves both metrics in 12 settings and trades one metric against the other in three. MLP improves both metrics in 12 settings and has a trade-off in two. The robust-density baseline improves both metrics over MLP only for Open3DSG at $K = 5$.

Direct-verifier diagnostics. Hard-tail and Hard-drop read primary-verifier labels at evaluation time and are therefore non-deployable diagnostics, not comparable learned baselines. Hard-tail often lowers Violation further but trades Recall against the learned estimators. Hard-drop guarantees zero primary Violation by construction, but it returns fewer than K rows for Open3DSG at $K \in \{50, 100\}$ and for VL-SAT and SGFN at $K = 100$. The complete selected-count and uncertainty rows are provided in the machine-readable results.

Compatibility estimators and matched fusion. The MLP is a matched nonlinear estimator rather than a separate method. RankAvg and reciprocal rank fusion (RRF) use the Linear compatibility ranks:

$$\mathrm{RankAvg}_i = \tfrac{1}{2}\left(\tilde{r}_i^Z + \tilde{r}_i^C\right), \qquad \mathrm{RRF}_i = \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \tag{1}$$

Here r_i^Z and r_i^C are within-family ranks, and tildes denote normalized ranks. Main Table 1 reports both fusion rules at every K. Their predictor-dependent Recall–Violation trade-offs show that the result is not explained by replacing the product with a uniformly stronger fusion rule.

Matched structural controls. Table 11 reports the matched structural controls for the MLP estimator. Main Table 2 reports the corresponding Linear controls.

Across both reported K values and both compatibility estimators, corrupted predicates and pair geometry increase verifier-derived Violation, while compatibility-only ordering loses substantial Recall on VL-SAT and SGFN. Open3DSG remains more sensitive to source-score scale.

Pooled compatibility ablation. A single pooled Linear head tests whether family-specific estimation matters. At $K = 50$, pooling changes Recall by $+0.07$, $+0.33$, and $+2.21$ percentage points for VL-SAT, Open3DSG, and SGFN, respectively, but increases Violation by 0.16, 2.38, and 0.34 points. The result is a predictor-dependent trade-off rather than a uniformly stronger estimator. Full $K = 100$ rows are provided in the machine-readable results.

Routing-constraint controls. The *family-aware* rule preserves the source family label at every rank position. It reranks proximity and vertical-order candidates in separate family lists. The matched *joint P/V* control uses the same candidates, compatibility estimates, product scores, and support/contact positions and identities. It differs only by placing proximity and vertical-order candidates in one list. Table 12 reports this comparison at all five K values.

No rule is best in every setting. For Open3DSG, Joint P/V raises Linear Recall at all five K values. For the MLP, it raises Recall at $K = 5, 10$ but lowers Recall by 2.92 and 3.80 percentage points at $K = 50, 100$, with both paired intervals below zero. At Open3DSG $K = 50$, the MLP control changes the selected proximity/vertical-order counts from 6,295/6,508 to 3,423/9,380 while leaving all 13,833 support/contact selections unchanged. We therefore interpret family-aware re-ranking as a constraint that preserves relation-family composition, not as a rule that maximizes aggregate performance.

Two broader relaxations are retained as scope comparisons. *Support order only* preserves the relative source order of support/contact candidates but permits competition across families. *All-family product* also applies learned compatibility to support/contact. These conditions change support/contact positions or selected membership and are not matched tests of the family-aware constraint. Complete rows are included in the machine-readable results.

Open3DSG coverage sensitivity. The public preprocessing drops a context when fewer than four annotated objects

Predictor	Ranking	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Source	0.93	3.61	7.27	16.43	26.00
	Linear	0.51	3.06	6.36	14.34	21.50
	MLP	0.42	2.79	6.17	14.05	21.57
Open3DSG	Source	83.20	65.18	53.80	45.96	40.87
	Linear	1.26	1.60	2.90	4.90	8.65
	MLP	16.84	13.18	9.17	8.84	12.32
SGFN	Source	0.00	2.42	5.33	11.56	21.55
	Linear	0.00	1.54	3.89	8.13	15.98
	MLP	0.00	1.32	3.55	7.63	15.88

Table 8: Point- and mesh-based consistency audit. Agreement-based Violation (%, ↓) on the shared 3DSSG validation target. A satisfied or violated label requires agreement between the point- and mesh-based measurements.

		Five smooth mappings			Percentile mapping		
Predictor	Estimator	Both	Min. ΔR	Max. ΔV	Both	Min. ΔR	Max. ΔV
VL-SAT	Linear	25/25	0.000	-0.036	5/5	+0.025	-0.146
	MLP	24/25	-0.025	-0.109	3/5	-0.201	-0.146
Open3DSG	Linear	25/25	+0.302	-9.163	5/5	+0.277	-8.982
	MLP	25/25	+0.277	-8.240	5/5	+0.201	-7.851
SGFN	Linear	25/25	0.000	0.000	3/5	-0.227	0.000
	MLP	25/25	0.000	0.000	3/5	-0.050	0.000

Table 9: Source-score mapping sensitivity over $K \in \{5, 10, 20, 50, 100\}$. Both counts settings with $\Delta R \geq 0$ and $\Delta V \leq 0$ relative to Source. Minimum Recall and maximum Violation changes are percentage points. Negative ΔV is preferred. The smooth block pools three power and two logit-temperature mappings.

retain view metadata. This accounts for all 15 missing contexts. The main evaluation retains all 548 contexts and the 3,972-relation denominator, assigning no predictions to the missing contexts. At $K = 100$, Source and Linear obtain 51.11/12.42 and 56.85/3.24 Recall/Violation percentages. Recovering predictions for the missing contexts changes these values only to 51.61/12.42 and 57.35/3.32. The direction of the result is therefore stable, while the main paper retains the conservative public route.

Candidate-Pool Oracle Recall. RelCompat3D re-ranks a fixed candidate pool and cannot recover a ground-truth relation that the source did not generate. We quantify this boundary using three post-hoc upper bounds. The family-aware oracle preserves the source top-K family counts and support/contact subsequence. The family-count oracle preserves only family counts. The unconstrained oracle selects up to K exact-match candidates without family constraints. None of these oracles is a deployable comparator or a model-selection criterion.

Table 13 summarizes candidate-pool coverage and the three Recall upper bounds.

Open3DSG has a substantially lower candidate-pool coverage (79.68%) than the two closed-set sources (99.72%). Even the family-aware oracle remains above the best active variant for every predictor, so the current method does not saturate the attainable re-ranking ceiling. Full results for all five K values are provided in the machine-readable results.

Paired Scan-Level Intervals. Paired 95% intervals use 1,000 scan-level bootstrap resamples. All contexts from a sampled scan are resampled together, and the same resample indices are used for Source and each RelCompat3D variant.

Table 14 reports the paired $K = 50$ changes used in the main-paper interval interpretation.

For Open3DSG and SGFN, both intervals exclude zero in the favorable direction for both metrics. For VL-SAT, the Recall intervals contain zero while the Violation intervals remain below zero. Complete intervals for all five K values are provided in the machine-readable results.

External Transfer Stress Test. We apply the 3DSSG-trained compatibility model and family-aware re-ranking rule to the official ReplicaSSG test scenes using FROSS predictions. No target-specific parameter fitting, normalization, or imputation is performed in this evaluation. The exact mapping covers `near`, `above`, and `under`. Support/contact is excluded because the ontologies do not provide an exact mapping. We report this as a transfer stress test rather than evidence of dataset-level generalization.

Table 15 reports this stress test at all five K values.

Paired scene-bootstrap intervals support joint improvement at $K = 10$ (ΔR [.0048,.1467], ΔV [$-$.1000,$-$.0091]) and $K = 50$ (ΔR [.0179,.0811], ΔV [$-$.0683,$-$.0185]). The $K = 20$ Recall interval touches zero, while $K = 5$ is inconclusive. At $K = 100$, heavy quantization of the source relation score leaves little room for multiplicative re-ranking. Candidate coverage also limits exact-match Recall

Predictor	Ranking rule	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Source	41.94/0.29	63.22/0.82	80.74/1.42	92.72/2.68	96.35/4.76
	Robust density	41.14/0.33	62.13/0.97	79.83/1.51	91.77/2.68	96.10/4.46
	Linear	42.07/0.15	63.39/0.57	80.82/1.14	92.77/1.97	96.58/2.95
	MLP	42.09/0.15	63.47/0.51	80.92/1.09	92.72/1.89	96.50/2.96
Open3DSG	Source	3.42/52.05	9.87/32.89	19.89/20.99	40.43/13.87	51.11/12.42
	Robust density	4.03/2.03	11.56/3.45	22.61/4.30	43.76/5.09	56.92/6.05
	Linear	3.73/0.94	11.38/2.33	23.62/3.13	44.18/3.42	56.85/3.24
	MLP	3.70/4.95	11.78/4.97	24.67/4.56	46.70/4.13	59.89/3.71
SGFN	Source	31.17/2.37	39.75/3.49	49.12/3.22	74.02/3.85	92.35/6.30
	Robust density	31.14/2.37	39.43/3.58	48.51/3.31	73.87/4.01	91.84/6.19
	Linear	31.17/2.37	39.75/3.47	49.14/2.97	74.50/2.63	93.03/3.50
	MLP	31.17/2.37	39.75/3.47	49.19/2.96	74.57/2.58	92.88/3.50

Table 10: Comparison with the non-learned robust-density baseline on the shared 3DSSG validation target. Cells are exact-match Recall / verifier-derived Violation percentages. Linear and MLP denote the two proposed RelCompat3D estimators under the same family-aware re-ranking rule.

to 44.19%. The stress test therefore supports K-dependent transfer, not dataset-level generalization.

Family Composition. Table 16 reports the slice for each family inside the actual global top-100 ranking. Deltas compare RelCompat3D-Linear with Source using shared paired context-bootstrap indices.

Support/contact is exactly unchanged because the ranking procedure preserves its selected subsequence and the source family-label sequence at every K. Vertical order drives most of the aggregate reduction. The result is consequently a scoped improvement, not a uniform family claim. The same direction holds across the reported K values. Full machine-readable family slices and simultaneous intervals accompany the code package.

Verifier-Uncertainty Sensitivity. The reported $\mathrm{V@K}_{all}$ includes uncertain candidates in its denominator but does not label them satisfied. We therefore recompute each ranking with $\mathrm{V@K}_{dec}$ over satisfied/violated candidates, uncertainty rate $U@K$, and an uncertain-as-violation variant $\mathrm{V@K}_{u\to v}$ that counts every uncertain candidate as a violation. Table 17 gives the $K = 100$ comparison.

RelCompat3D-Linear-minus-Source changes in V_{dec} are $-.0277$ for VL-SAT, $-.1578$ for Open3DSG, and $-.0440$ for SGFN. Their paired 95% intervals all exclude zero. Uncertainty-rate changes are +.0009, $-.0076$, and +.0071. Uncertain-as-violation changes are $-.0173$, $-.0993$, and $-.0209$. Both decidable-only and uncertain-as-violation V are lower for all three predictors, although uncertainty does not decrease uniformly. Across both estimators, primary, decidable-only, and uncertain-as-violation V are non-increasing relative to Source in all 30 predictor–K settings. The Hard-drop diagnostic has zero Violation when the denominator includes all verifier statuses, but it retains uncertain candidates and therefore has uncertain-as-violation V of .3712/.4722/.4042. Thus zero measured Violation does not imply reliable or complete selection.

Machine-readable result rows. The accompanying code and data package contains the full per-K, per-seed, per-mapping, interval, selected-count, and coverage rows summarized in this document, together with manifests and tolerance checks for the reported paper cells.

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	MLP full	46.70	4.13	59.89	3.71
	Wrong predicate	46.78	19.30	58.89	18.34
	Wrong pair	38.02	9.31	49.90	8.90
	Shuffled geometry	36.83	12.88	47.94	12.38
	Fixed-predicate swap	45.52	19.13	57.43	18.15
	Compatibility only	48.19	3.42	61.33	3.28
VL-SAT	MLP full	92.72	1.89	96.50	2.96
	Wrong predicate	90.99	5.03	94.99	8.31
	Wrong pair	91.31	2.62	95.62	4.71
	Shuffled geometry	90.26	3.00	94.71	5.49
	Fixed-predicate swap	90.99	4.99	95.09	8.13
	Compatibility only	77.84	1.61	90.18	2.05
SGFN	MLP full	74.57	2.58	92.88	3.50
	Wrong predicate	71.78	9.67	90.01	13.52
	Wrong pair	72.10	3.91	89.05	6.59
	Shuffled geometry	71.15	4.59	87.84	8.06
	Fixed-predicate swap	71.85	9.65	90.23	13.35
	Compatibility only	57.10	2.33	80.99	2.31

Table 11: Matched controls for RelCompat3D-MLP. Recall (R, ↑) and verifier-derived Violation (V, ↓) on the shared 3DSSG validation target. All values are percentages. Main Table 2 reports the corresponding Linear controls.

Predictor	Estimator	Re-ranking	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Linear	Family-aware	42.07/0.15	63.39/0.57	80.82/1.14	92.77/1.97	96.58/2.95
	Linear	Joint P/V	42.04/0.15	63.42/0.55	80.84/1.11	92.72/1.95	96.60/2.94
	MLP	Family-aware	42.09/0.15	63.47/0.51	80.92/1.09	92.72/1.89	96.50/2.96
	MLP	Joint P/V	42.12/0.15	63.42/0.49	80.97/1.06	92.77/1.86	96.53/2.93
Open3DSG	Linear	Family-aware	3.73/0.94	11.38/2.33	23.62/3.13	44.18/3.42	56.85/3.24
	Linear	Joint P/V	7.98/0.94	15.84/2.33	25.33/3.13	44.76/3.39	57.83/3.08
	MLP	Family-aware	3.70/4.95	11.78/4.97	24.67/4.56	46.70/4.13	59.89/3.71
	MLP	Joint P/V	7.48/4.17	14.85/4.77	23.92/4.66	43.78/4.16	56.09/3.68
SGFN	Linear	Family-aware	31.17/2.37	39.75/3.47	49.14/2.97	74.50/2.63	93.03/3.50
	Linear	Joint P/V	31.17/2.37	39.75/3.47	49.19/2.97	74.67/2.61	93.28/3.39
	MLP	Family-aware	31.17/2.37	39.75/3.47	49.19/2.96	74.57/2.58	92.88/3.50
	MLP	Joint P/V	31.17/2.37	39.75/3.47	49.27/2.96	75.25/2.57	93.13/3.37

Table 12: Matched routing-constraint comparison on the shared 3DSSG validation target. Cells are exact-match Recall / verifier-derived Violation percentages. Joint P/V changes only the competition between proximity and vertical-order candidates. Support/contact positions and identities remain fixed.

Predictor	Pool cov.	Source@50	Best@50	Fam.-aware@50	Fam.-count@50	Unconstr.@50	Fam.-aware@100
VL-SAT	99.72	92.72	92.77	96.73	99.52	99.72	98.79
Open3DSG	79.68	40.43	46.70	63.72	73.46	79.68	71.88
SGFN	99.72	74.02	74.57	86.05	88.07	99.72	98.49

Table 13: Candidate-pool Recall upper bounds. Exact-match Recall (%) on the shared 3DSSG validation target. Pool cov. is the fraction of ground-truth relations present in the fixed candidate pool. Best is the better active RelCompat3D variant at $K = 50$.

Estimator	Predictor	ΔR@50	ΔV@50
Linear	VL-SAT	$+0.05$ $[-0.05, +0.18]$	-0.70 $[-0.88, -0.55]$
	Open3DSG	$+3.75$ $[+2.96, +4.50]$	-10.45 $[-11.06, -9.86]$
	SGFN	$+0.48$ $[+0.29, +0.69]$	-1.22 $[-1.44, -1.01]$
MLP	VL-SAT	0.00 $[-0.10, +0.10]$	-0.79 $[-0.98, -0.62]$
	Open3DSG	$+6.27$ $[+5.02, +7.49]$	-9.74 $[-10.31, -9.13]$
	SGFN	$+0.55$ $[+0.32, +0.81]$	-1.27 $[-1.51, -1.06]$

Table 14: Paired scan-level changes at $K = 50$. Each cell reports the RelCompat3D-minus-Source change and its paired 95% interval in percentage points. Positive Recall and negative Violation changes are favorable.

K	Source R/V	RelCompat3D-Linear R/V	ΔR/ΔV
5	.0465/.0545	.0698/.0182	+.0233/−.0364
10	.0756/.0818	.1453/.0273	+.0698/−.0545
20	.1453/.1273	.2209/.0364	+.0756/−.0909
50	.2616/.1328	.3140/.0904	+.0523/−.0424
100	.3547/.1967	.3547/.1958	+.0000/−.0010

Table 15: ReplicaSSG/FROSS transfer across all reported K values. R is exact-match Recall over 172 relations. V is verifier-derived Violation. The Linear estimator and family-aware re-ranking procedure match the main experiment.

Predictor	Family	ΔR	ΔV
VL-SAT	Support/contact	.0000	.0000
VL-SAT	Proximity	+.0040	−.0076
VL-SAT	Vertical order	+.0051	−.0609
Open3DSG	Support/contact	.0000	.0000
Open3DSG	Proximity	+.0968	−.0443
Open3DSG	Vertical order	+.1538	−.3014
SGFN	Support/contact	.0000	.0000
SGFN	Proximity	+.0142	−.0150
SGFN	Vertical order	+.0051	−.0643

Table 16: Family-specific metrics within the selected top-100 predictions. Positive ΔV is a regression. V is verifier-derived Violation.

Predictor	Condition	V_{all}	V_{dec}	U	$V_{u \to v}$
VL-SAT	Source	.0476	.0729	.3464	.3941
VL-SAT	Linear	.0295	.0452	.3473	.3768
Open3DSG	Source	.1242	.2126	.4161	.5402
Open3DSG	Linear	.0324	.0548	.4085	.4409
SGFN	Source	.0630	.1005	.3732	.4362
SGFN	Linear	.0350	.0565	.3803	.4153

Table 17: Uncertainty sensitivity at $K = 100$. Lower is better for all four reported quantities.